

# Dose-Aware MR-to-CT Synthesis through a Differentiable Physics-Prior Residual 3D U-Net for Photon Dose Prediction on 0.35 T MRI (DoseRAD2026 Photon-MRI Task)

Kai Wang, Meixu Chen, and Rui Yang

Department of Radiation Oncology, University of Colorado Anschutz Medical Campus, Aurora, CO, USA

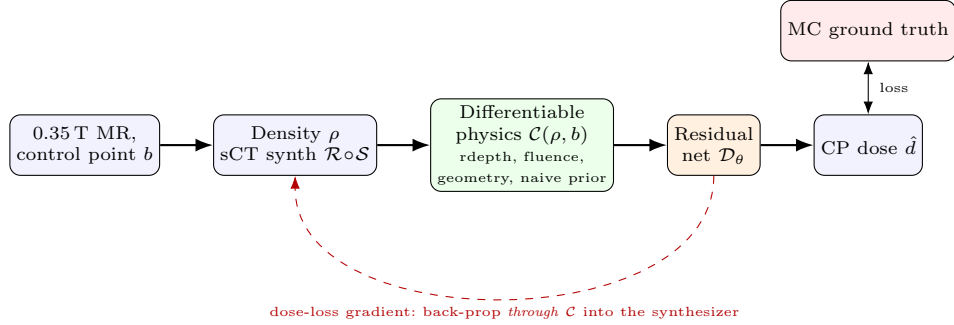
{kai.2.wang,meixu.chen,ray.yang}@cuanschutz.edu

**Abstract.** We predict photon control-point dose directly from 0.35 T MRI by making synthetic-CT (sCT) generation *dose-aware*. A classifier-refiner synthesizer maps the MR to a density image; an analytical, differentiable physics operator turns density and control point into six dose-shaped channels; and a residual 3D U-Net corrects the analytical prior toward Monte Carlo. Because the physics is differentiable in the density, the synthesizer is trained *jointly through it* with the dose loss; the sCT is optimized for the dose, not for image fidelity. This matters: across five coarse-prior variants, standalone image quality did not predict fused dose accuracy ( $r=0.02$ ), and a TotalSegmentator-MRI lung mask with far better Dice than our classifier gained nothing dosimetrically. The full chain lifts  $\gamma 1\%/1\text{ mm}$  from 68.7 (bulk-density) through 82.6 (image-fidelity sCT) and 88.8 (dose-aware synthesis) to 90.7 (joint training) on the development fold; 5-fold cross-validation gives  $91.1 \pm 1.6$ , and a cross-validated real-CT ceiling of  $95.8 \pm 1.3$  isolates the remaining synthesis cost at  $4.7 \pm 1.4$  points. On the hidden test set, MR intensity shift between training and test images emerged as the dominant failure mode and was substantially mitigated by retraining the tissue classifier with strong intensity-shift augmentation. The final full-data submission scored  $\gamma = 84.11$ , per-beam MAE 0.0172, IDD 0.0127, DVH 1.00 at 94.8 s per job; the fastest runtime on the preliminary leaderboard at submission time.

**Keywords:** MR-only planning · synthetic CT · dose-aware synthesis · differentiable physics · MR-linac

## 1 Introduction

On MR-linacs and in MR-only workflows the planning image is an MRI, whose intensities encode neither electron density nor attenuation. The established remedy (synthesize a CT, then compute dose) optimizes the synthesis for an *image* objective (MAE, SSIM, Dice) that is only a surrogate for the clinically relevant quantity [1,2,3,4,5,6]. We instead couple synthesis and dose in one differentiable



**Fig. 1.** Dose-aware MR-only pipeline: the synthesizer is trained jointly through the differentiable physics with the dose loss (dashed), so the sCT serves the dose, not an image metric.

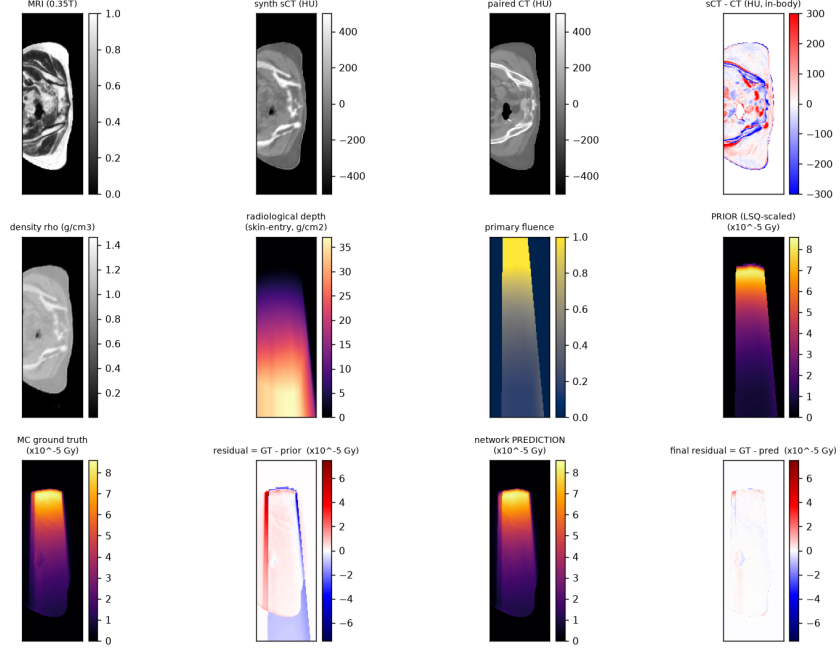
pipeline. An analytical physics operator  $\mathcal{C}$  maps density and control-point descriptor to dose-shaped channels; a residual 3D U-Net  $\mathcal{D}_\theta$  corrects the analytical prior toward Monte Carlo (MC); and because  $\mathcal{C}$  is differentiable in the density, a synthesizer  $\mathcal{X} = \mathcal{R} \circ \mathcal{S}$  placed upstream is trained *through the physics* with the dose loss. The synthesizer is thereby free to place density where the *dose* needs it rather than where an image metric would reward it; and our central empirical finding is that these two objectives genuinely diverge (Sec. 3.3). This report details the Photon-MRI system; the same framework serves our submissions to all four DoseRAD2026 tasks.

## 2 Methods

### 2.1 Data

Official DoseRAD2026 training set [10] (75 patients: 39 thorax, 36 abdomen; SynthRAD2025 cohort [11]): per patient a co-registered 0.35 T MR (ViewRay MRIdian, bSSFP, breath-hold) and planning CT, beam parameters, and per-control-point MC dose-to-medium (matRad [9]) on a  $2 \times 2 \times 2$  mm grid; 540 control points per patient. Per the challenge definition, the MRI-task ground truth is the CT-computed MC dose mapped onto the co-registered MR, so the achievable target for MR-only calculation is the CT dose itself; the framing behind our real-CT-ceiling analysis (Sec. 3.3). The sCT refiner additionally uses 166 field-matched 0.35 T MR–CT pairs from the public SynthRAD2025 pool (disjoint from validation patients); no other external data. Development: 5-fold CV over all 75; **the final submission is retrained on all 75**, with the leaderboard as held-out validation.

Photon-MRI (skin-entry, MR->sCT->dose) - 1ABB006 CP 0\_134 (abdomen) | prior<->GT corr 0.899 pred rel-MAE(>5%) 2.8%

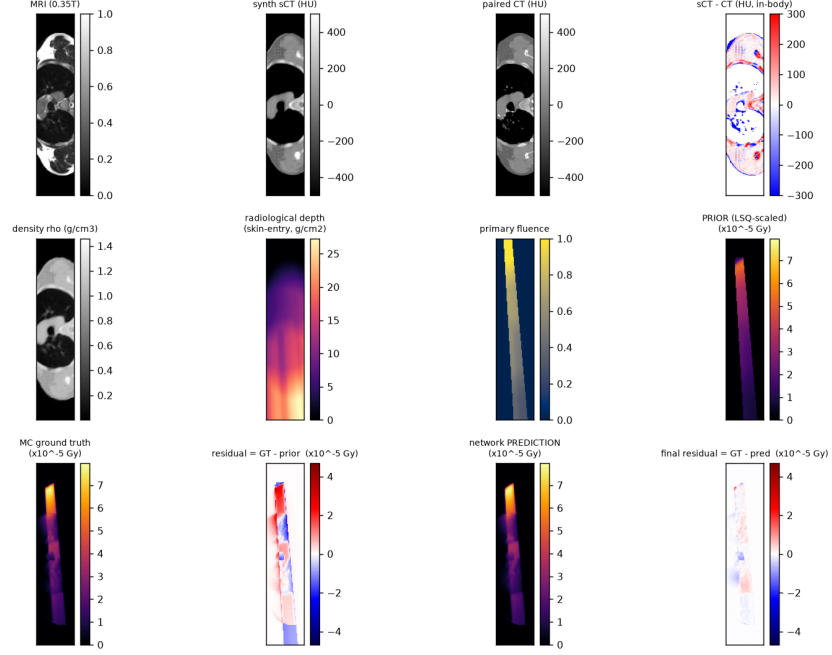


**Fig. 2.** Deployed Photon-MRI chain (abdomen, 1ABB006). Row 1: synthesis ; MR (0.35 T)  $\rightarrow$  synthetic CT  $\rightarrow$  paired real CT  $\rightarrow$  in-body sCT-CT residual. Remaining panels: physics channels on the synthetic density  $\rightarrow$  LSQ-scaled naive prior  $\rightarrow$  MC ground truth  $\rightarrow$  residual (GT-prior)  $\rightarrow$  prediction  $\rightarrow$  final residual (GT-pred).

## 2.2 Model

**Synthesizer**  $\mathcal{K} = \mathcal{R} \circ \mathcal{S}$ :  $\mathcal{S}$ , a whole-image 3D U-Net classifier, assigns each voxel to air/lung/soft-tissue/bone (CT-HU band edges  $-500/-300/+200$ ) and maps classes to a bulk-HU coarse CT;  $\mathcal{R}$ , a whole-image refiner, maps [MR, coarse] to the sCT, from which the official HU calibration yields density. **Physics**  $\mathcal{C}$ : the six photon channels (density, ray-marched radiological depth, MLC-projected primary fluence, distance-to-CAX, source distance, naive analytical surrogate), all differentiable tensor operations, with the per-ray skin-entry rule (depth accumulates from the first  $\rho > 0.05 \text{ g/cm}^3$  crossing; prior masked strictly upstream). **Dose network**  $\mathcal{D}_\theta$ : DoseUNet3D (base 48; four levels; ASPP bottleneck; Soft-plus head; FiLM modality conditioning), warm-started from the Photon-CT network. Figure 2 shows the full deployed chain: synthesis quality and its dosimetric consequence in one figure.

Photon-MRI (skin-entry, MR->sCT->dose) - 1THB048 CP 2\_133 (lung) | prior->GT corr 0.837 pred rel-MAE(>5%) 6.2%



**Fig. 3.** As Fig. 2, lung case: 0.35 T MR carries little lung density information; lung is the site of the largest synthesis cost.

### 2.3 Training

Stage 1: the tissue classifier and refiner are pretrained (the refiner also on the 166 field-matched pairs); the dose network is warm-started from Photon-CT. Stage 2 (dose-aware): refiner and dose network train *jointly*, 15–20k steps, with the dose loss back-propagating through  $\mathcal{C}$  into the refiner, plus a region-weighted per-voxel sCT anchor to the real CT as regularizer. Loss and optimization as in our CT system (high-dose-weighted  $L_1$ , AdamW, EMA, patches  $128^3$ ). **Robustness for the hidden test:** the dominant held-out failure mode was MR *intensity shift* between our training MRs and the hidden-test MRs (diagnosed by leaderboard probes: internally synthesized shifts reproduced the drop, and the shared tissue classifier accounted for most of it). The deployed classifier is therefore retrained with strong MR intensity-shift augmentation (bias/gamma/scale perturbations); no augmentation is used in dose training itself. Reproducibility: fixed seeds, YAML configs, EMA checkpoints; code and weights to be open-sourced (Sec. 6).

**Table 1.** Photon-MRI results. CV: 5-fold over the 75 public patients. Hidden test: official metrics of the final full-data submission (August 2026). Gate: 181 s/job.

| Metric                                    | Value                                                                     |
|-------------------------------------------|---------------------------------------------------------------------------|
| CV plan $\gamma 1\%/1\text{ mm}$ (5-fold) | $91.1 \pm 1.6$                                                            |
| CV real-CT ceiling / synthesis cost       | $95.8 \pm 1.3$ / $4.7 \pm 1.4$                                            |
| Hidden-test per-beam masked MAE           | $0.0172 \pm 0.0060$                                                       |
| Hidden-test IDD curve distance            | $0.0127 \pm 0.0079$                                                       |
| Hidden-test stratified plan MAE           | $0.0129 \pm 0.0070$                                                       |
| Hidden-test plan $\gamma 1\%/1\text{ mm}$ | $84.11 \pm 7.73$                                                          |
| Hidden-test DVH clinical score            | $1.00 \pm 0.77$                                                           |
| Hidden-test runtime per job               | 94.8 s (fastest on the board at submission; sCT $\sim 2\text{ s/image}$ ) |

## 2.4 Evaluation

Internal: plan-level local  $\gamma$  [8] (pymedphys,  $1\%/1\text{ mm}$ ,  $\geq 10\%$  Rx cutoff, byte-identical to the vendored official code), per-beam masked MAE, official IDD and stratified-MAE reimplementations; a frozen 16-patient held-out cohort gates every deployment change (*no accuracy loss, no speed loss*). Mechanistic probes: the *real-CT ceiling* (the trained MRI pipeline re-run with the real CT substituted for the sCT, isolating the synthesis cost patient-matched) and a five-variant coarse-prior sweep relating image fidelity to fused dose accuracy. Runtime: official  $T = t_{\text{fix}} + N \cdot t_{\text{dose map}}$ , gate 181 s, double-weighted.

## 3 Results

### 3.1 Accuracy: cross-validated and hidden-test

Per-fold CV  $\gamma 1/1$ : 90.7/90.2/92.2/88.9/93.6 (mean  $91.1 \pm 1.6$ ). Development fold ( $n = 16$ ):  $\gamma 1/1 = 90.7 \pm 5.7$  (abdomen 94.6, lung 86.8),  $\gamma 2/2 = 97.7$ ,  $\gamma 3/3 = 99.1$ ; per-beam masked MAE  $1.55 \pm 0.38\%$ . Table 1 adds the hidden-test metrics. The CV-to-hidden-test gamma drop ( $91.1 \rightarrow 84.1$ ) is dominated by MR intensity shift (Sec. 2.3): on the sister proton-MRI task the shift-augmented classifier recovered +3.1 board gamma, and the same fix is deployed here.

### 3.2 Ablation: each dose-aware stage pays

Table 2: bulk-density assignment through the engine reaches 68.7; a learned image-fidelity sCT 82.6; making the synthesis dose-aware (frozen engine) 88.8; joint training of synthesizer and dose network 90.7. The two learned-synthesis stages are complementary; the coarse prior lifts lung, dose-aware training lifts abdomen.

**Table 2.** Photon-MRI ablation (plan  $\gamma$ 1%/1 mm, development fold,  $n = 16$ ).

| configuration                                     | ALL         | abd  | lung |
|---------------------------------------------------|-------------|------|------|
| bulk-density coarse CT $\rightarrow$ engine       | 68.7        | 80.4 | 56.9 |
| learned sCT (image-fidelity) $\rightarrow$ engine | 82.6        | 89.5 | 75.7 |
| + dose-aware synthesis (frozen engine)            | 88.8        | 93.7 | 84.0 |
| + joint dose-net training (deployed)              | <b>90.7</b> | 94.6 | 86.8 |

### 3.3 Image fidelity does not predict dose accuracy

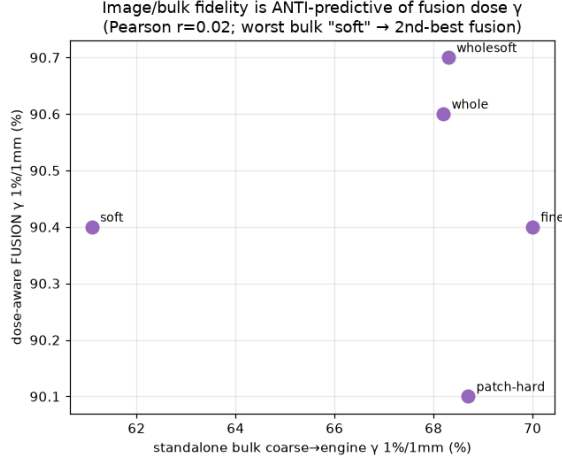
Across five coarse-prior variants passed through the dose-aware fusion, standalone bulk- $\gamma$  was uncorrelated with fused dose  $\gamma$  (Pearson  $r=0.02$ ; Fig. 4). The extremes make the point concretely: the *worst* standalone variant (a soft-tissue-only bulk map, standalone  $\gamma$  61.1) gave the second-best fusion result (90.4), while the variant with the *lowest* classifier Dice achieved the best fusion (90.7). A TotalSegmentator-MRI [7] lung mask (Dice 0.79 vs. our classifier’s 0.33) did not improve dose. The mechanism is exactly the joint training: gradients from the dose loss place density where the dose needs it, independently of coarse quality. This directly cautions against selecting sCT models by MAE/SSIM/Dice for dosimetric use. The cross-validated real-CT ceiling ( $95.8 \pm 1.3$ ) bounds what any synthesis can add: the remaining photon synthesis cost is  $4.7 \pm 1.4$  points, roughly half the corresponding proton cost ( $8.7 \pm 1.9$ ), because photon dose is density-tolerant while proton range integrates density (Fig. 5).

### 3.4 Diagnosing the hidden-test drop: MR intensity shift

The gap between cross-validated (91.1) and hidden-test (84.1) gamma is itself a result worth dissecting, because it is the failure mode every clinical MR-only deployment will face. Three observations triangulate it. First, synthetically shifting our validation MRs (bias/gamma/scale perturbations at magnitudes estimated from the test-set image statistics we could observe) reproduced a drop of the observed size, and attributing it stage-by-stage placed most of it in the shared tissue *classifier*, not the refiner or dose network. Second, retraining the classifier with strong intensity-shift augmentation recovered a substantial part of the board drop on the sister proton-MRI task (+3.1 board gamma) and is deployed here. Third, the recovery is partial and our synthetic shifts were harsher than the real ones, a calibration gap we report so others can budget for it: internal shift probes over-predicted the recoverable gain by roughly  $4\times$ . The residual hidden-test gap after augmentation is consistent with a mixture of residual intensity shift and true anatomical distribution shift in the larger test cohort.

### 3.5 Runtime engineering

Runtime is double-weighted, and this submission held the fastest runtime on the preliminary board at submission time (94.8 s; the compressed-output variant of

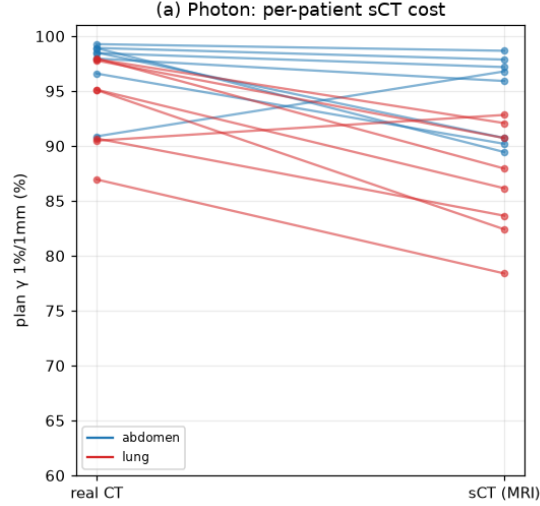


**Fig. 4.** Image fidelity does not predict dose accuracy. Each point is one coarse-prior variant of the synthesis chain: *soft* (soft-tissue-only bulk map), *whole* (whole-image classifier), *wholesoft* (whole-image classifier, soft-tissue bulk values), *fine* (finer tissue classes), *patch-hard* (patch-based classifier, hard labels). Horizontal axis: the variant’s *standalone* quality, the plan  $\gamma$  1%/1 mm obtained when its bulk coarse CT is fed directly to the dose engine. Vertical axis: the *fusion* accuracy, the plan  $\gamma$  of the full dose-aware pipeline built on that variant (jointly trained synthesizer + dose network). Standalone quality spans 9 points but fusion accuracy varies by only 0.6 (Pearson  $r=0.02$ ): the jointly trained refiner recovers dose-relevant density regardless of the coarse prior.

the same model ran 124.7 s, and margin 24 added a further  $\sim 30$  s; both rejected by measurement). Photon-MRI adds one patient-level stage to the Photon-CT path; the synthesizer runs once per source image ( $\sim 2$  s, of which the two whole-image resamples and two network forwards are profiled separately in the container log); after which the per-CP loop is identical: aperture-bbox crops (margin 16, selected on full-grid metrics), GPU-resident channels, compiled dose net, cutoff-zeroed uncompressed output (photon jobs are compute-bound; an earlier compressed variant was 30 s slower). Two accuracy-relevant deployment details: the naive prior at deploy is the same skin-entry-gated surrogate the network was trained with (the CT container’s build-up variant is *not* interchangeable), and the whole-image classifier requires the same inference mode as training, a sliding-window variant collapsed lung on some grids in the sister proton container, a failure mode we gate for explicitly.

## 4 Discussion

The central lesson of this task is that *dose-aware synthesis works and image metrics mislead*:  $+8.1 \gamma$  from bulk to learned sCT,  $+6.2$  more from making synthesis serve the dose (Table 2), while image fidelity was non-predictive of the outcome

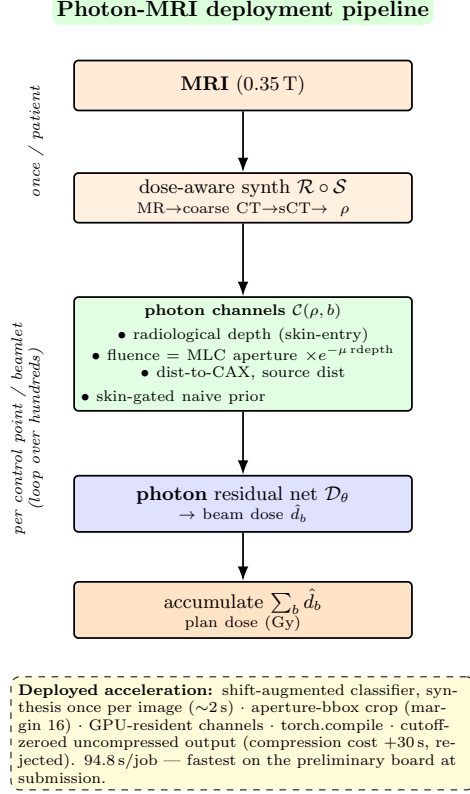


**Fig. 5.** Per-patient synthesis cost on this task (real CT vs. sCT input, development fold; blue abdomen, red lung): photon drops are small and *mixed*; photon dose is density-tolerant. The corresponding proton cost is  $\sim 2\times$  larger (see text).

( $r=0.02$ ). The remaining gap to the real-CT ceiling (4.7 points cross-validated) is a genuine synthesis cost, concentrated in lung at 0.35 T where the MR simply carries little density information; better segmentation and robustness augmentation both failed to close it, suggesting a data limit (field strength, cohort size) rather than a capacity limit. **Limitations and generalizability.** Single benchmark, single 0.35 T scanner; the hidden-test intensity shift we diagnosed is a preview of the cross-device shift any clinical deployment faces, and our mitigation (shift-augmented classification) recovers only part of it. The ground truth is the CT-computed dose mapped to the MR, so residual CT–MR registration error is label noise. MR-linac photon planning with bulk or sCT densities is already clinically accepted; our contribution is a stronger, dose-optimized MR-to-dose path and a caution about how to select its components.

*Why an intermediate sCT at all?* Three design questions deserve explicit answers. First, the intermediate density image exists because it is the *interface to the physics*: the analytical operator (radiological depth, fluence attenuation) consumes a density map, and it is this operator that supplies the residual-learning advantage and the data efficiency of the whole framework; a direct MR-to-dose network would have to relearn that physics from 75 patients. Second, the synthesizer is partially transferred: the refiner is pretrained on 166 field-matched SynthRAD2025 MR–CT pairs before dose-aware joint training. Third, the dose-aware sCT genuinely *diverges* from an imaging-optimized sCT: trained through the dose loss, the synthesizer accepts a worse image error where the dose does not need fidelity and spends its capacity where the dose does; this is precisely



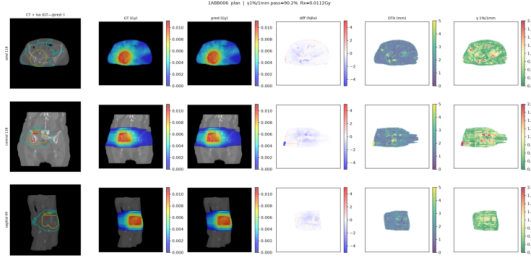


**Fig. 6.** The deployed Photon-MRI container: per patient, the density stage runs once; the differentiable-channels  $\rightarrow$  residual-net loop runs per control point/beamlet. The callout lists the deployed acceleration measures and their measured effects.

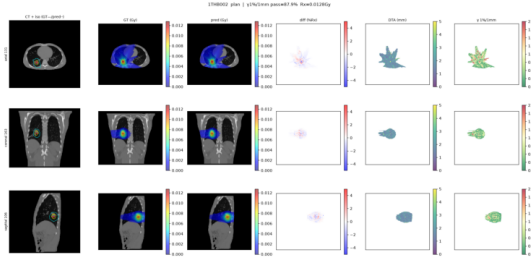
why image fidelity stops predicting dose accuracy (Sec. 3.3), and it means a dose-aware sCT should not be reused as a diagnostic-quality image. Whether a fully sCT-free direct MR-to-dose model can match this pipeline remains open (we did not run that controlled experiment) and is our first future-work item.

**Future work.** (i) *sCT-free direct MR-to-dose*: the fidelity result implies the sCT is only a means to an end; a network consuming the MR (plus geometry channels computed on a provisional density) end-to-end may bypass the synthesis bottleneck entirely, and the strongest competing entries in this challenge appear to point the same way. (ii) *Shift-robust synthesis by design*: replace heuristic intensity augmentation with test-time intensity-normalization or physics-based bSSFP signal modelling, targeting the classifier that our attribution identified as the shift-sensitive stage. (iii) *Speed*: our distilled base-32 photon student ( $2.1\times$ , teacher-exceeding on CT) transfers to this task only after finetuning on synthesis-domain channels, a domain-adapted student is trained and, if it passes our no-

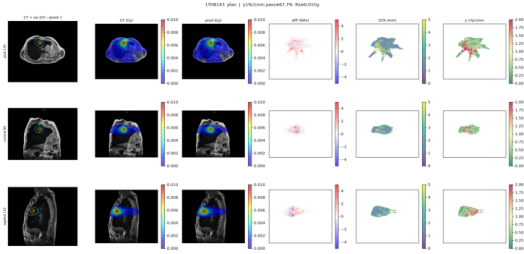
**Abdomen (1ABB006) — plan  $\gamma 1\%/1\text{mm} = 90.2\%$**



**Lung (1THB002) — plan  $\gamma 1\%/1\text{mm} = 87.9\%$**



**Hard case (1THB143) — plan  $\gamma 1\%/1\text{mm} = 67.7\%$**



**Fig. 7.** Qualitative Photon-MRI results on three cases spanning the difficulty range: abdomen (1ABB006, plan  $\gamma 1/1 = 90.2$ ), lung (1THB002, 87.9), and the geometrically hard case 1THB143 (67.7; the worst patient in all four challenge tasks simultaneously, yet largely passing at 3%/3 mm: small-magnitude errors at steep lung–tissue interfaces). Columns: image with GT (solid) and predicted (dashed) isodose, MC dose, predicted dose, difference, distance-to-agreement, local  $\gamma 1\%/1\text{ mm}$ .

accuracy-loss gate, halves the per-CP cost. (iv) *Higher field and larger cohorts*: the lung synthesis cost tracks the information limit of 0.35 T bSSFP; 1.5 T MR-linac data is the clean test of how much of the remaining 4.7-point ceiling gap is physics rather than data.

## 5 Author Contributions (CRediT)

**Kai Wang:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Meixu**

**Chen:** Software (deployment and runtime optimization), Investigation (acceleration study), Validation, Writing – review & editing. **Rui Yang:** Conceptualization, Methodology, Writing – review & editing, Supervision.

## 6 Other Information

**Funding:** No external funding was received for this work. **Collaborators:** None beyond the listed authors. **Writing assistance:** Large-language-model assistance (Claude, Anthropic) was used in drafting and editing the text of this report; all methods, experiments, analyses and conclusions are the authors’ own. **Code availability:** The full implementation (models, differentiable physics operators, deployment containers, and weights) will be released open-source at <https://github.com/wangkaiwan/PhyPriorNet> before the challenge deadline (2026-10-01).

## References

1. Tian, L., et al.: MRI-based proton dose calculation for pelvic tumors using deep learning. *Phys. Med. Biol.* **71**, 045003 (2026)
2. Gao, Y., et al.: MRI-only based material mass density and relative stopping power estimation via deep learning for proton therapy. *Sci. Rep.* **14**, 11166 (2024)
3. Liu, Y., et al.: MRI-based treatment planning for proton radiotherapy: dosimetric validation of a deep-learning liver synthetic-CT method. *Phys. Med. Biol.* **64**, 145015 (2019)
4. Kazemifar, S., et al.: Dosimetric evaluation of synthetic CT generated with GANs for MRI-only proton therapy planning of brain tumors. *J. Appl. Clin. Med. Phys.* **21**(5), 76–86 (2020)
5. Li, X., et al.: Uncertainty-aware MR-based CT synthesis for robust proton therapy planning of brain tumour. *Radiother. Oncol.* **189**, 109902 (2023)
6. Liu, R., et al.: Synthetic dual-energy CT for MRI-only based proton therapy treatment planning using label-GAN. *Phys. Med. Biol.* **66**, 065014 (2021)
7. Wasserthal, J., et al.: TotalSegmentator MRI: robust sequence-independent segmentation of anatomic structures in MRI. *Radiology* (2025)
8. Low, D.A., Harms, W.B., Mutic, S., Purdy, J.A.: A technique for the quantitative evaluation of dose distributions. *Med. Phys.* **25**, 656–661 (1998)
9. Wieser, H.P., et al.: Development of the open-source dose calculation and optimization toolkit matRad. *Med. Phys.* **44**, 2556–2568 (2017)
10. DoseRAD2026 Grand Challenge, <https://doserad2026.grand-challenge.org> (2026)
11. Thummerer, A., et al.: SynthRAD2025 Grand Challenge dataset: generating synthetic CTs for radiotherapy. *Med. Phys.* (2025)